

Chemical reactions:

Summary

Student: Class:

Complete the statements below to compile a summary of this unit. The missing words can be found in the word list below. Two of the words may be used twice.

- In a chemical reaction, the atoms of the are rearranged and new bonds are formed. In this way, new chemicals, the, are formed.
- The Law of Conservation of Mass states that matter can be neither nor during a chemical reaction.
- It is important to know the correct of the reactants and products when writing a balanced chemical equation.
- An solution is one in which a substance is dissolved in water.
- When two aqueous solutions react, a solid sometimes appears.
- All compounds containing the sodium ion, the ammonium ion, the ion and the ion will dissolve in water.
- Redox reactions involve the of electrons from one atom, ion or molecule to another.
- Redox reactions are actually two reactions, and, happening simultaneously.
- In the 'dry' of metal, electrons are transferred from metal atoms to oxygen molecules.
- A reaction occurs when one element displaces another element from a compound.
- Hydrocarbons are chemicals that contain only and
- When one compound breaks down into two or more simpler chemicals, a reaction has occurred.
- Magnesium reacts readily with acid, but gold does not. This is due to the different of the metals.
- When metals are reacted with hydrochloric acid, gas is produced.
- A is produced when an acid and a base react together in a neutralisation reaction.
- A base containing the carbonate ion will produce as well as a salt and water.
- A is a substance that speeds up a chemical reaction without reacting in the reaction.
- Amylase is an example of a biological catalyst, or
- The production of light by a chemical reaction is called

transfer

dioxide

potassium

enzyme

formulae

catalyst

salt

oxidation

products

hydrogen

carbon

reactivity

precipitate

reactants

created

corrosion

displacement

destroyed

decomposition

nitrate

reduction

aqueous

electrons

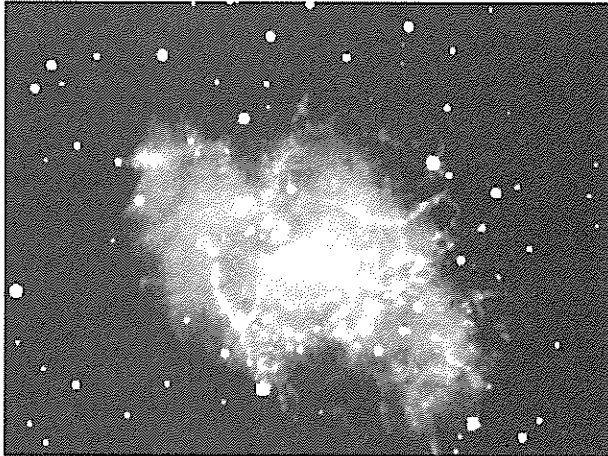
chemiluminescence

Observing stars

Student: Class:

1. Three types of celestial objects are illustrated below. Identify and outline the nature of each.

(a)



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Outline:

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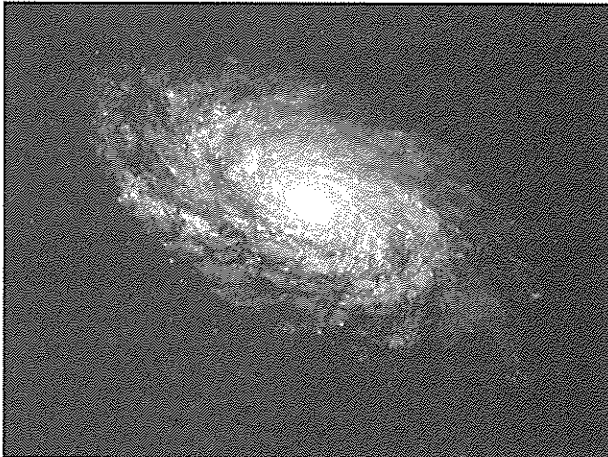
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(b)



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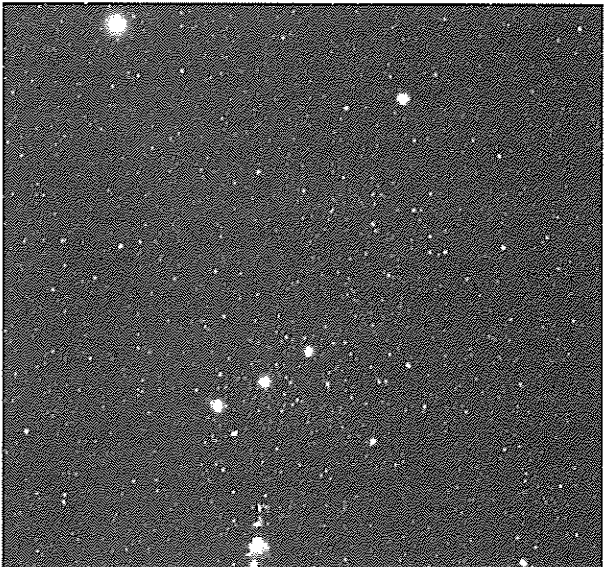
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(c)



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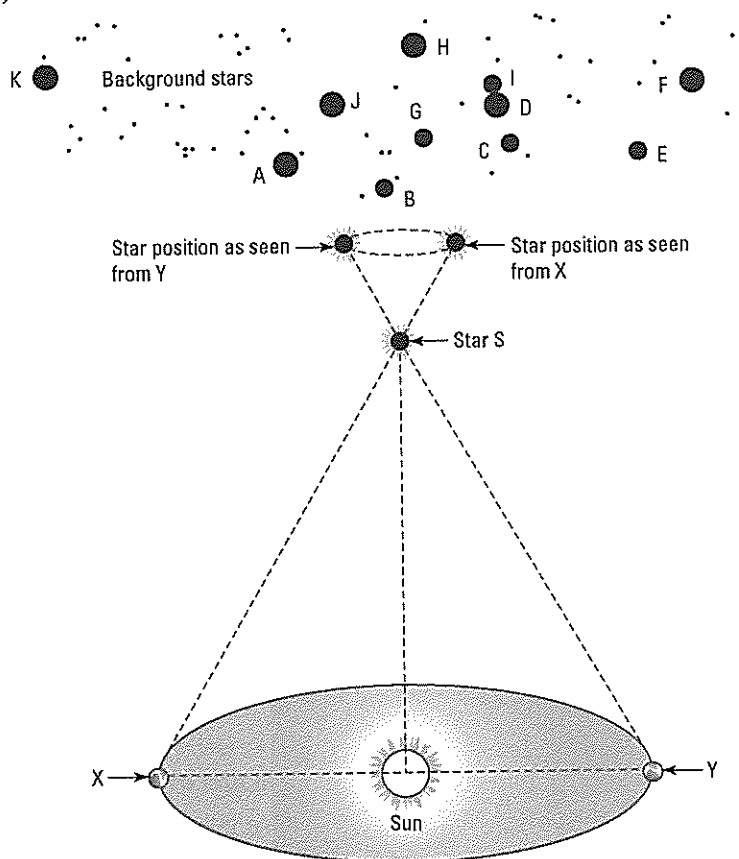
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2. The following diagram shows the effect of parallax on the observation of a close star (S) from two locations (X and Y) in the Earth's orbit around the Sun.



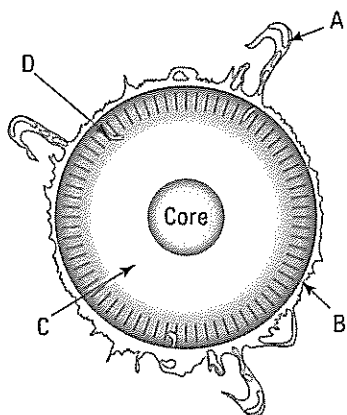
- (a) The Earth is shown in two positions X and Y. How long does it take for the Earth to move from X to Y?
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- (b) Explain the meaning of parallax in relation to the apparent motion of stars.
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- (c) Before 1837 it was believed that all stars existed on a transparent celestial sphere at a fixed distance from the Earth. How did the observation of parallax of close stars change this view?
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- (d) Outline the motion of the stars as seen by an observer on Earth during the time taken to travel from X to Y.
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The sun and nuclear fusion

Student: Class:

The structure of the sun

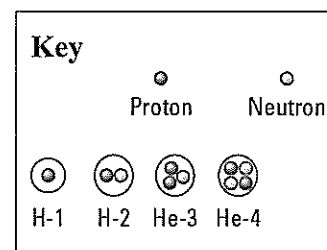
1. Our sun has a layered structure. Use the diagram and table below to match the letters (A–D) to the correct structure.



Description of structure	Letter
1. Radiation zone — this layer is outside the core; energy transfers to this layer from the core	
2. Solar prominence — these structures extend up to 500 000 km into space	
3. Convection zone — convection currents carry energy towards the photosphere	
4. Photosphere — light is emitted into space from this surface	

Nuclear fusion

2. The sun generates energy by nuclear fusion. During this process hydrogen nuclei are joined together to form helium nuclei. This process occurs in three steps. Use the key to construct diagrams for each step. The first step has been done for you.



Step 1: Two hydrogen (H-1) nuclei fuse to form a deuterium (H-2) nucleus.



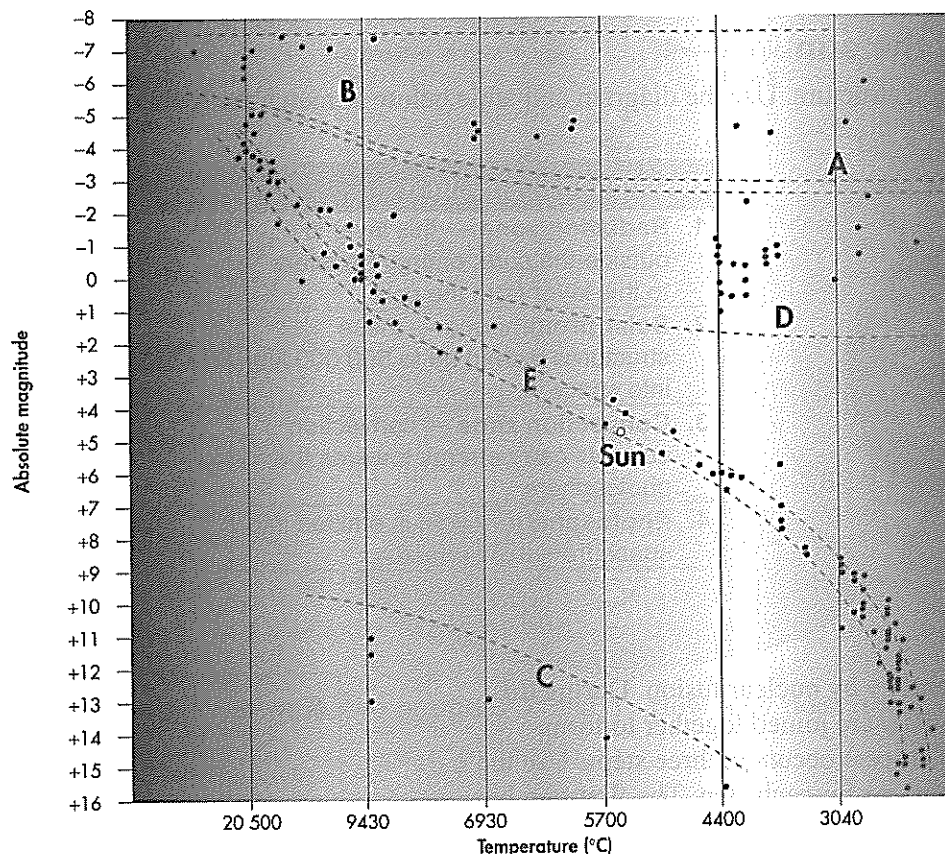
Step 2: One deuterium nucleus and one hydrogen nucleus fuse to form a helium-3 (He-3) nucleus. A gamma ray is also formed.

Step 3: Two helium-3 nuclei fuse to form a helium-4 (He-4) nucleus. Two protons and large amounts of energy are released in this step.

The brightness of stars

Student: Class:

The Hertzsprung–Russell diagram is a plot of the temperature of the surface of the star against the brightness of stars (absolute magnitude).

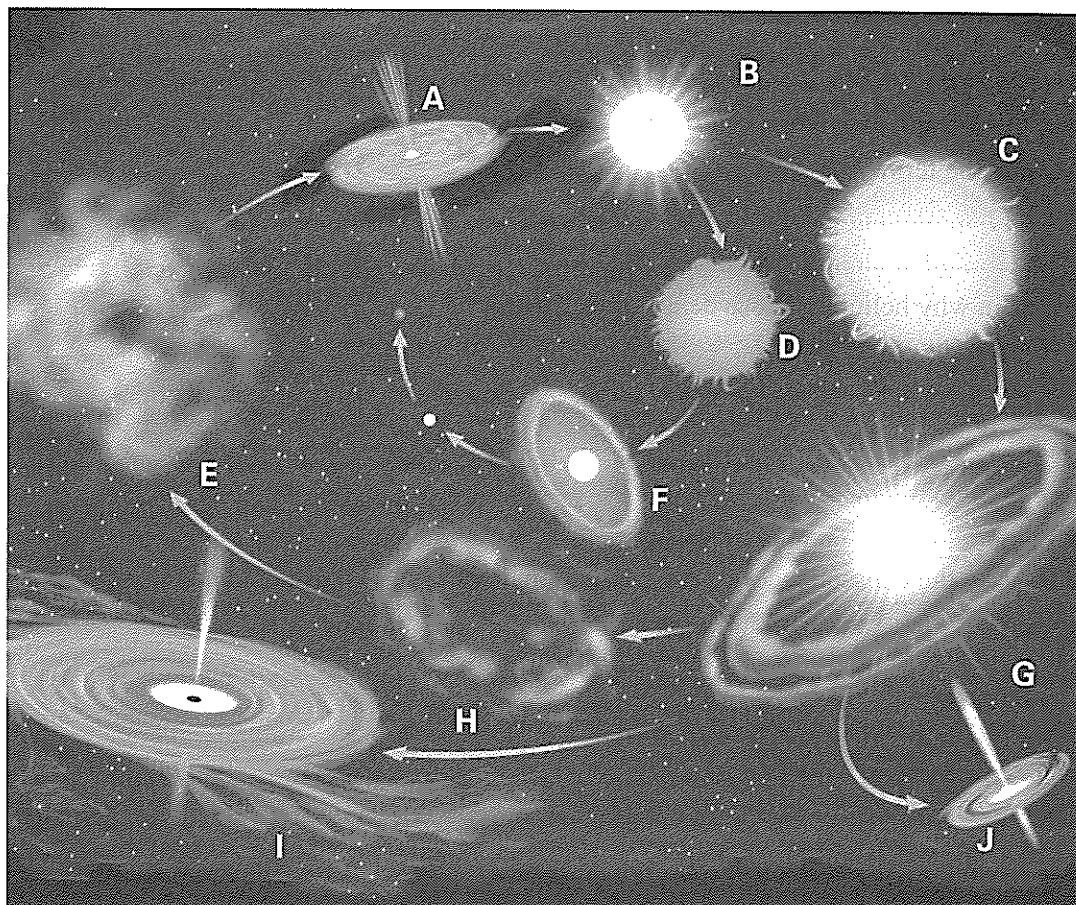


- Where are the brightest stars located in this graph?
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- Where are the dimmest, coolest stars located in the graph?
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- Identify the type of star represented by the letters:
A:
B:
C:
D:
E:
- During the next stage in our sun's life cycle it will move off the main sequence of the graph. Which region of the graph (A–E) will it move to next?
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- Explain how the absolute magnitude of a star is different to its apparent magnitude.
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- A star has an absolute magnitude of -6 and a surface temperature of 2900°C .
(a) Predict the colour of this star.
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(b) Identify the group that this star belongs to.
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Star life cycles

Student: Class:

1. The following diagram shows the evolutionary sequence of stars of different masses. Match each of the images, A–J, with the events listed in the table below. These events are randomly listed.



Event	Letter
Star forms and begins to shine	
Black hole forms from a red super giant explosion (supernova)	
Pulsar (neutron star) forms	
Red giant star forms from a main sequence star	
White dwarf forms at the core of a red giant star	
Supernova formed from a red super giant	
Protostar forms	
Red super giant forms from stars heavier than our sun	
Large nebula forms as gravity pulls dust and gases together	
Small planetary nebula formed after supernova event	

2. Read the following text and answer the questions below.

RED GIANTS

Stars that have a mass between one solar mass and eight solar masses will evolve into red giant stars. Betelgeuse in the Orion constellation is a red giant star. During their lives, stars like our sun, fuse hydrogen atoms together to form helium. Since helium is denser than hydrogen, the helium that forms sinks to the core of the star. The star 'burns' the hydrogen in the shell that surrounds this core. When the amount of helium in the star reaches about 12% the core collapses inwards under gravity leading to intense heating. The heat causes the outer layers of the star to expand outwards. The star can swell to one hundred times its original size. The surface of these giant stars begins to cool so that the star now shines with a red light.

- (a) Define the term 'one solar mass'.

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- (b) Explain why helium is more dense than hydrogen.

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- (c) Describe the consequences of the gravitational collapse of the star's core.

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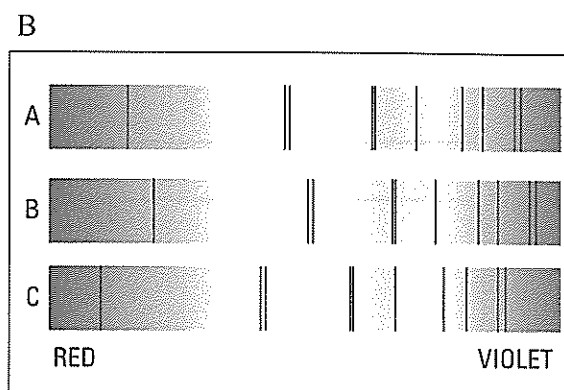
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Student: Class:

1. Figure A shows a light emitting object moving to the right. The light emitted is detected by two stationary observers (O_1 and O_2). Figure B shows absorption spectra of a light emitting object. The dark lines represent light colours that have been absorbed by different elements in the star.

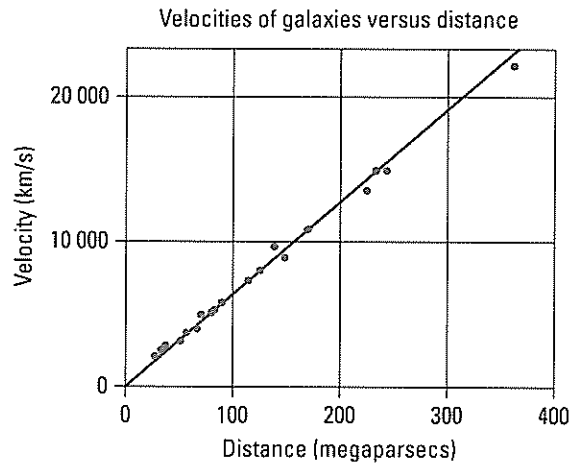


- [illegible]

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Hubble's expanding universe

2. The following graph shows a plot of the velocities of a number of galaxies moving away from our galaxy versus their distance from the Earth.



- (a) Describe the trend demonstrated by the graph.

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- (b) Explain why these galaxies are moving away from our own.

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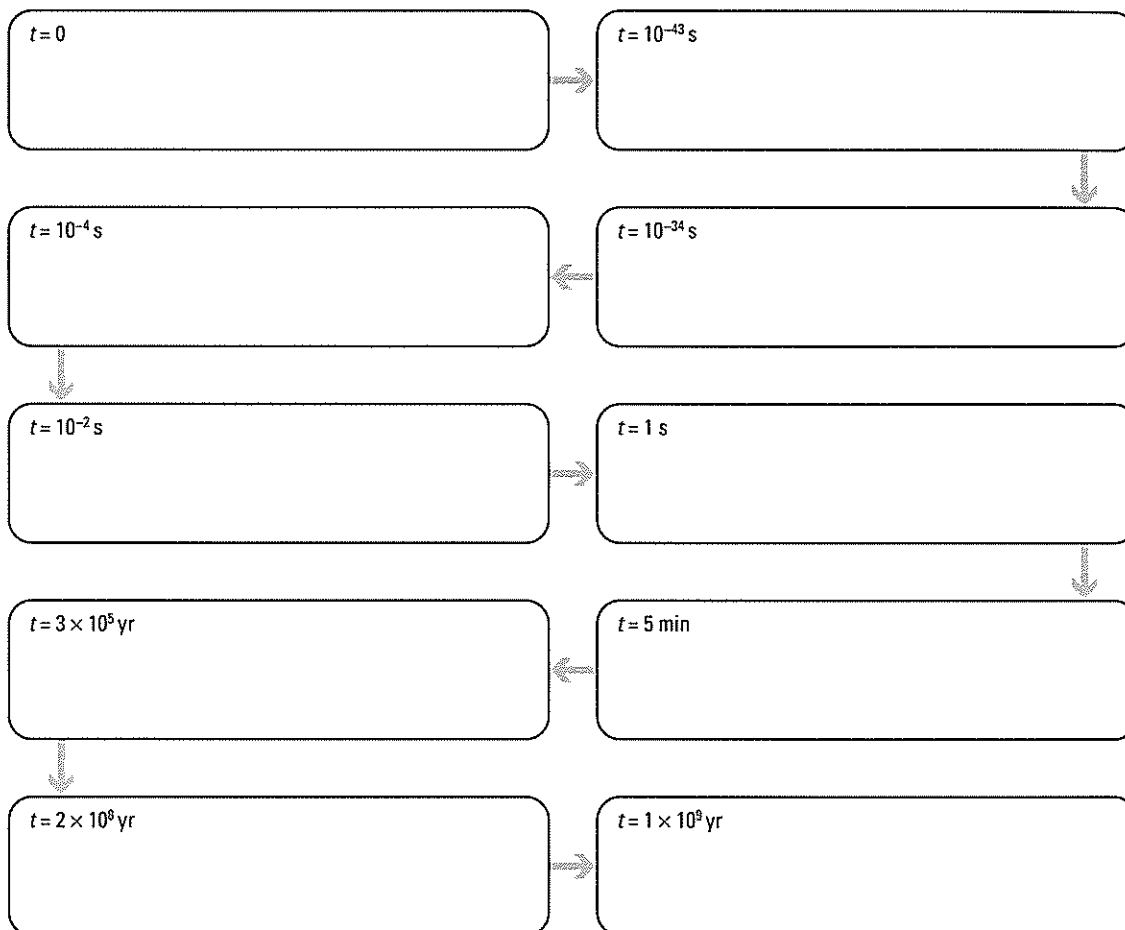
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The big bang

Student: Class:

Flowcharting the big bang

1. Complete the big bang flow chart by selecting the event from the list below that matches each step in the sequence and rewrite it in the correct box. Enter a title for each box that summarises each event.

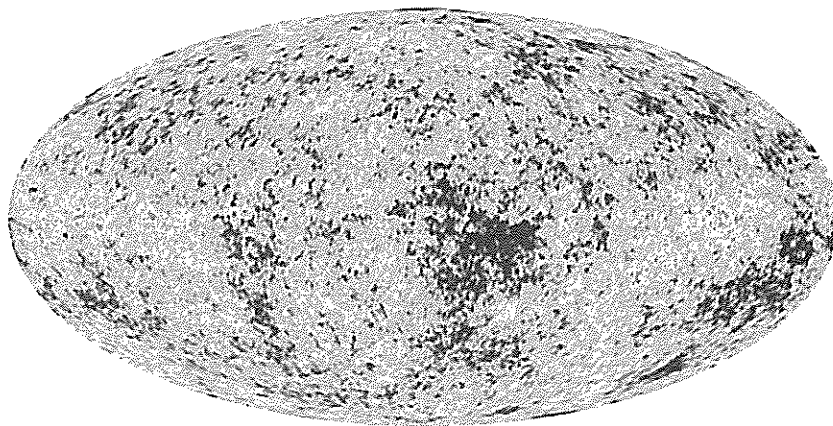


Big bang events in random order:

- A. Universe has cooled to 3000 °C; first atoms form
- B. Galaxies begin to form
- C. Time and space form; space expands rapidly
- D. Heavier matter forms — protons, neutrons
- E. The big bang — singularity forms
- F. Universe cools; reaches size of the solar system
- G. Atomic nuclei form
- H. First stars appear
- I. Light matter forms — electrons, positrons
- J. Universe has cooled to 10¹⁰ °C

Evidence for the big bang

2. The following microwave map of the early universe ($t = 380\,000$ years after big bang) was produced by a probe that orbited the Earth in 2001.



- (a) The map shows that in the early universe matter was not uniformly distributed. Explain how this map supports the big bang theory.

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- (b) In 1948, Gamow and Alpher predicted that microwave radiation should now fill the universe. They proposed that this radiation was the cooled 'afterglow' of the big bang. When was this microwave radiation first detected?

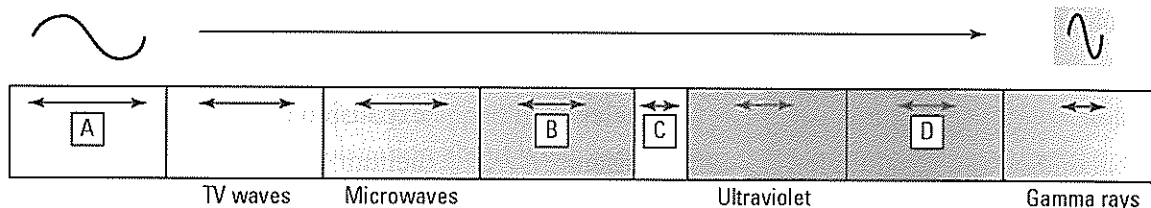
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Telescopes

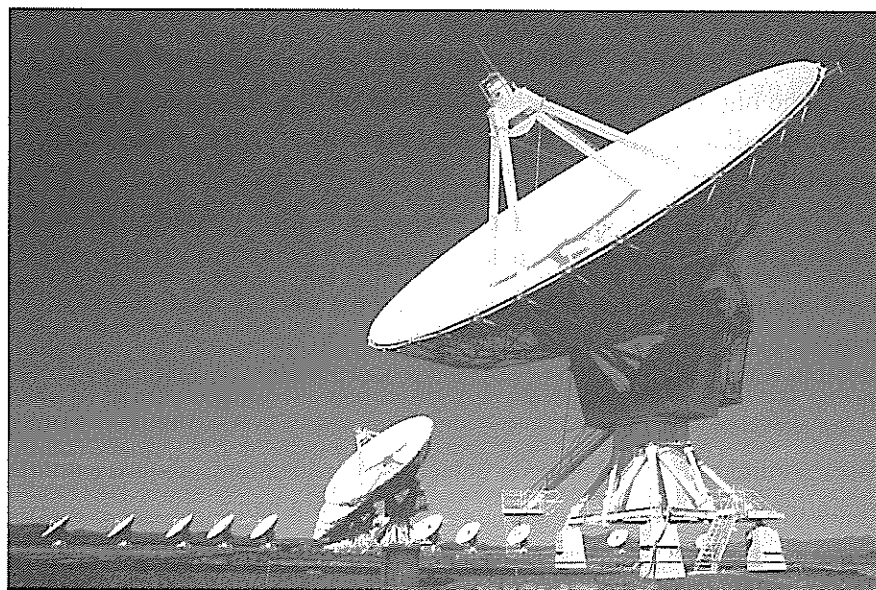
Student: Class:

1. Astronomers use the electromagnetic spectrum to observe the universe. The diagram below shows this spectrum. Four of the regions are labelled A, B, C and D.

- (a) Identify the region (A–D) of the spectrum that is used by optical telescopes to explore the universe.



- (b) The photograph below shows another type of telescope. Name this type of telescope and identify the region of the spectrum (A–D) that this type of telescope uses to observe the universe.



2. Complete the following cloze passage.

Optical (or visible light) telescopes can only examine the sky night. They should also be located well away from large cities which produce significant pollution. Radio telescopes can be used at any time of the or night. Radio waves have the ability to travel through the vast distances of space and our atmosphere significant distortion. Radio astronomy allows scientists to observe cooler interstellar gas clouds which are regions in which new are formed. Some astronomical objects emit very little visible light they often emit large numbers of radio waves. telescopes can therefore be used to observe them. Infra-red telescopes are also used in astronomy. They are placed onboard as the atmosphere absorbs too many infra-red waves. These infra-red telescopes have helped scientists to observe the afterglow of the and thus provide support for that theory.

The mysterious universe: Summary

Student: Class:

Use the listed words to complete the sentences.

absolute	protostar	electrons	spiral	observer
constellations	space	giants	billion	radio
fusion	time	nuclei	expansion	temperature
nebulae	afterglow	pulsars	helium	

- Groups of stars form patterns that are called
- Our solar system is part of a large galaxy called the Milky Way.
- Clouds of dust and gas in interstellar space are called
- The apparent movement of stars at different distances is caused by the movement of the This effect is called parallax.
- According to the big bang theory, the universe is about 13.7 years old.
- The first stage of the big bang involved the formation of and space. The early universe was very small and very hot. Space then underwent rapid and as it expanded it cooled.
- Quarks and were the first matter particles to form. Subsequent cooling led to proton and neutron formation which then combined to form
- Red shifting of lines in star spectra is consistent with the stretching of as the universe expands.
- The big bang theory can explain the large amount of relative to hydrogen in the universe.
- Microwave background radiation has been detected by instruments aboard satellites. This microwave radiation is the cooled of the big bang.
- Stars are formed when gravity causes gas and dust to come together and heat up. Eventually a is formed and then finally a star of a given mass.
- Stars like our sun evolve into red before they explode.
- Red supergiants evolve into black holes or
- The magnitude of a star is a measure of its real brightness when all stars are compared at the same distance.
- The Hertzsprung–Russell diagram plots the surface of the star versus its absolute magnitude. This diagram helps us understand the evolutionary life cycle of a star.
- Stars generate their energy by nuclear in which hydrogen nuclei join together to form heavier nuclei.
- The universe can be studied using optical (visible light) telescopes as well as telescopes.

Nature's time machine

Student: Class:

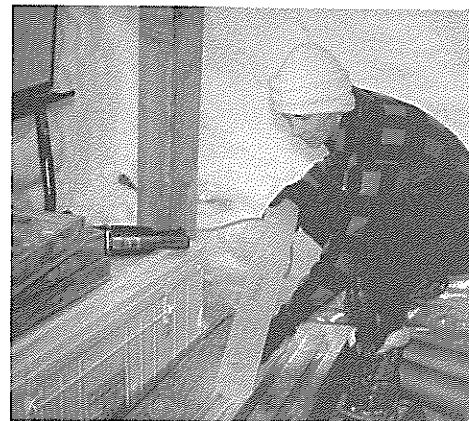
Secrets in the ice

Scientists may not have discovered a time machine yet, but they do have a way of looking back over 300 000 years. They do this by looking at **ice cores**. Scientists drill into the ice of Antarctica and collect samples from deep down. Ice cores provide information about the Earth and the atmosphere over hundreds of thousands of years.

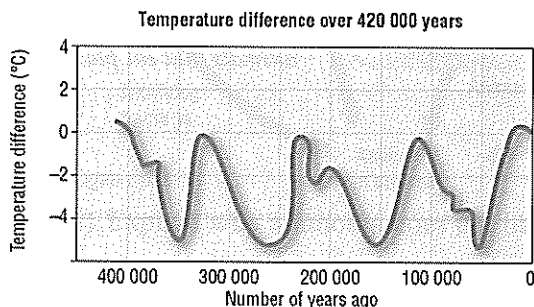
For thousands of years, snow has fallen in Antarctica. The snow turns to ice, which builds up over time. Dust, gases and other substances from the air become trapped in the ice. The trapped substances provide information about what was in the air at the time the snow fell.

By studying dust in the ice cores, scientists have been able to work out when certain volcanoes have erupted. They have linked radioactive substances in the ice to nuclear disasters in various parts of the world. The dust and radioactive substances were probably carried to Antarctica by the wind.

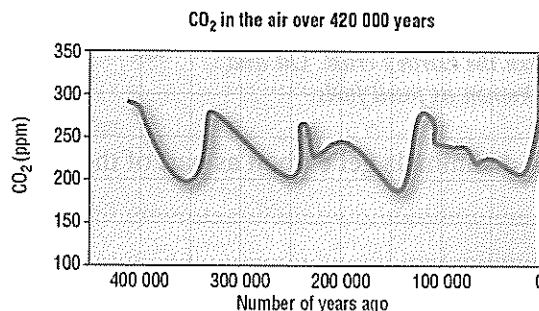
Scientists have used ice cores to track the amount of carbon dioxide in the air and the Earth's temperature. The graphs on this page show how these have changed over the past 400 000 years.



➤ This ice core was drilled from more than 3.7 km down. Parts of it are more than 150 000 years old.



Zero on the temperature scale represents the Earth's average temperature now. Greater than zero is warmer than now, and below zero is cooler. On the temperature line, the peaks represent warmer periods of time. In between the peaks, the average air temperature was up to 6°C lower than now. These colder times are called the ice ages.



The line on the graph for carbon dioxide resembles the temperature line. When the temperature was higher, there was more carbon dioxide in the air. When the temperature dropped, so did the amount of carbon dioxide. (The abbreviation ppm means parts per million.)

Look at the graphs above of how carbon dioxide and temperature have changed over time and answer the following questions.

- How many ice ages have there been in the past 400 000 years?
- At a point 350 000 years ago:
 - how much carbon dioxide was in the air
 - how much colder was the Earth's air compared to the present temperature?
- Has the Earth ever been warmer than it is now? If so, how much warmer?
- When was the amount of carbon dioxide in the air the highest?
- Why is an ice core like a time machine?